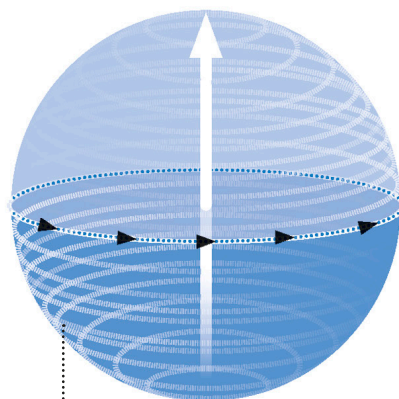


"SPIN"

In large-scale physics, spinning objects naturally possess a form of momentum due to a combination of their mass and rotation. Quantum particles possess a property (called intrinsic angular momentum, or spin) that was initially thought to be them spinning on the spot, but later turned out to be something much more curious. Spin does not refer to an actual physical rotation of particles, but shares many of the same characteristics as classical rotation and produces some similar effects.



The spin of a particle is measured against one of three dimensions, most often the dimension defined as the z-axis. When cast on this axis, the spin points in one of two directions: positive spin up or negative spin down.

SPIN UP

"But then the discovery of electron spin changed this picture considerably. The electron was not symmetrical. ... They are not simple, not so elementary as we had thought before."

Werner Heisenberg

SPIN DOWN

Rotating a "spin-up" $\frac{1}{2}$ electron through 360° results in its spin being reversed to "spin-down" or $-\frac{1}{2}$.

